Deep Neural Network Explainability Enhancement via Causality-Erasing SHAP Method for SAR Target Recognition

Zongyong Cui, *Member, IEEE*, Zhiyuan Yang, *Student Member, IEEE*, Zheng Zhou, *Graduate Student Member, IEEE*, Liqiang Mou, *Student Member, IEEE*, Kailing Tang, *Student Member, IEEE*, Zongjie Cao, *Member, IEEE*, and Jianyu Yang, *Member, IEEE*

***Abstract*— Deep neural networks (DNN)s have shown remarkable effectiveness in synthetic aperture radar (SAR) target recognition. However, the explainability problem for DNNs remains insufficiently addressed. One approach to tackle this challenge is the SHAP method. It enhances the explainability of DNNs in SAR target recognition by observing how the target, shadow, and clutter regions play their own distinct roles. The masked regions are typically filled with zero, mean, or random values in optical images. But if the same operation is performed on SAR images, it will affect the distribution of clutter and thus introducing new out-of-distribution (OOD) challenge. In this article, we propose a novel masking method to enhance the reliability and efficiency of the SHAP method in SAR-ATR applications. Experimental results on the MSTAR and OpenSARShip-1.0 datasets demonstrate that our proposed method provides a more faithful representation to show the importance of every single regions in SAR target recognition. Compared to methods using zero values, mean values, and random baselines, our proposed method significantly enhances the reliability of explainability.**

***Index Terms*— Baseline values, causality erasing, deep network explainability, SHAP, synthetic aperture radar (SAR) target recognition.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave sensor capable of all-weather and all-day imaging, unaffected by illumination and weather conditions. It holds significant value in both military and civilian applications for terrestrial observation. In recent years, deep neural networks (DNNs) have rapidly advanced in the field of remote sensing image analysis [1]. Since DNNs are used most for optical imaging processing, they have also been introduced to address SAR image processing tasks. Particularly, it has made significant strides in SAR image automatic target recognition (ATR) [2], garnering increasing attention.

Manuscript received 5 February 2024; revised 15 April 2024 and 21 May 2024; accepted 22 May 2024. Date of publication 27 May 2024; date of current version 10 June 2024. This work was supported by the National Natural Science Foundation of China under Grant 62271116. *(Corresponding author: Zongjie Cao.)*

The authors are with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: zycui@uestc.edu.cn; 202222010436@std.uestc.edu.cn; zhengzhou_uestc@163.com; 202222010438@std.uestc.edu.cn; 202222010437@std.uestc.edu.cn; zjcao@uestc.edu.cn; jyyang@uestc.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3405942

As deep learning has achieved remarkable success in SAR target recognition, the need for its explainability arises. One way to enhance the explainability of DNNs for SAR target recognition tasks is to observe the roles played by different regions.

Li et al. [3] quantitatively analyzed the degree of deep learning overfitting background clutter using Shapley values. Hosein Oveis et al. [4] employed the local interpretable model-agnostic explanations (LIMEs) method for incremental learning in SAR image analysis, enhancing human decision-making by providing auxiliary measurements alongside traditional classification scores. Fan et al. [5] employed the class activation mapping (CAM) method as an interpreter, allowing adaptive weight selection within the network to alleviate visual background noise in the interpreter's results.

Methods using DNNs to explain the role of different regions in the target recognition process is often called a post-hoc explainability method. This approach typically involves recognition before explaining networks. Common post-hoc explainability methods are listed in Table I. Actually, the terms explainability and interpretability are often used interchangeably, as there is no clear consensus on distinguishing between them. According to the authors' understanding, the distinction between the terms explainability and interpretability may be as follows: Interpretability concerns the ability of DNNs to produce understandable outputs and transparent predictions, aiding in mapping input–output spaces and understanding system boundaries; explainability focuses on the transparency of DNNs' internal workings, allowing for explanations of output results and revealing decision-making logic.

LRP [6] and DeepLIFT [7] are both methods used for explaining predictions in neural networks. LRP distributes relevance to input features to understand their contribution to the final decision, but it overlooks feature interactions. DeepLIFT decomposes neuron activations into linear combinations of input features, offering insights into individual computations, yet it can be complex and computationally demanding.

Belloni et al. [8] discussed the impact of different region arrangements on the recognition rates for each class, through an ablation study on target, clutter, and shadow regions. This method is relatively coarse and cannot quantitatively capture

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

the contributions of different regions. Li et al. [9] used the SHAP method for quantitative attribution of target, clutter, and shadow regions. They argued that background clutter attributes exhibit a false correlation with target classes, and deep learning may overfit clutter to reduce training errors. However, in this study, a normal distribution was employed as a baseline during masking operations. This method is proven ineffective in eliminating low-frequency information. Consequently, the contributions of different regions derived from their method may lack accuracy.

To accurately and faithfully reflect the impact of different regions on the overall outcome, this article proposes a method based on multivariate interactions to approximate the baseline values. The baseline values are then constrained by incorporating the distribution of SAR clutter to enhance their loyalty. When the baseline value has the capability to eliminate information from the original image without introducing additional information, it is referred to as satisfying loyalty. The contributions of this article can be summarized as follows.

1) AN SAR clutter distribution model is introduced to generate baseline values to fill the extracted region. This approach mitigates issues related to values exceeding the distribution range.
2) A method for acquiring baseline values is proposed based on multivariate interactions and clutter distribution. This method reduces salient patterns in the original samples while minimizing the introduction of new salient patterns. It ensures the reliability of SHAP method attributions.
3) Experimental validations on MSTAR and OpenSARShip-1.0 datasets demonstrate the effectiveness and optimal performance of the proposed method.

The remaining organization of this article is as follows. In Section II, relevant content of the SHAP method is briefly introduced, followed by the analysis of different methods for selecting baseline values and their limitations. Section III presents the overall framework of the proposed method, which gives a detailed explanation of the proposed method. The experimental setup, results, and corresponding analysis are described in Section IV. Finally, Section V summarizes the research findings of this article.

II. Preliminary Knowledge

To enhance the understanding of the SHAP method, this section provides preliminary knowledge from three perspectives: the Shapley values, bivariate Shapley interaction, and baseline values.

A. Shapley Values

The Shapley value is widely regarded as a strategy for assessing the overall contribution of each player in a game, originating from cooperative game theory. The Shapley value is frequently employed to estimate the attribution of input feature units in DNNs. Assuming $\boldsymbol{x}$ is the input of a DNN with n feature units. For instance, $\boldsymbol{x} = \{x_1, x_2, \ldots, x_n\}$, then the Shapley values for the ith feature unit $\Phi(i|\boldsymbol{n})$ can be defined as

$$\Phi(i|\boldsymbol{n}) = \sum_{s \subseteq \boldsymbol{n}\backslash\{i\}} \binom{n-1}{|\boldsymbol{s}|}^{-1} [f(\boldsymbol{x}_{\boldsymbol{s}\cup\{i\}}) - f(\boldsymbol{x}_{\boldsymbol{s}})] \tag{1}$$

where $f(\boldsymbol{x}_{\boldsymbol{s}})$ represents the model's output when the set of feature units $\boldsymbol{s}$ is present, with the feature units in $\boldsymbol{n}\backslash\boldsymbol{s}$ being masked. $\Phi(i|\boldsymbol{n})$ indicates the positive or negative impact of the ith feature unit on the outcome.

Consider an SAR image $I \in \mathbb{R}^{n\times n}$, where each pixel can be treated as a player. Each pixel can exist in two states, presence or absence, resulting in a total of $2^{n\times n}$ possibilities. For a certain pixel i, where $\boldsymbol{s} \subseteq \boldsymbol{n}\backslash\{i\}$, there are $n(n-1)$ possible combinations. Therefore, the computational complexity for calculating the Shapley value of pixel i is as follows:

$$\sum_{r=1}^{n^2-1} \binom{r}{n^2-1} = 2^{n^2-1}. \tag{2}$$

It is worth noting that the computational complexity of Shapley value is exponential, and directly optimizing (2) is NP-hard.

B. Bivariate Shapley Interaction

In practical scenarios, the shadow region responds to target shape information under specific sensor conditions [8]. This observation leads to the hypothesis that there might be a significant correlation between the targets and shadow regions in SAR images. To explore this relationship, the bivariate Shapley interaction is utilized to define this relationship, as proposed by Zhang et al. [15].

Given two players, i and j, if they always participate together in the game or always abstain simultaneously, players i and j can form a coalition. The contribution of this coalition to the game is often different from the sum of the contributions of i and j when they participate individually in the game. Consider the coalition formed by players i and j as a whole, a distinct new player represented by $\boldsymbol{s}_{ij} = \{i, j\}$.

For instance, in the case of SAR images, the target and shadow components can be considered as a coalition, comprising both the target and shadow segments. The Shapley value of the new player can also be calculated using (1). However, in this case, players i and j need to be excluded from the overall player set $\boldsymbol{n}$. Additionally, the new player $\boldsymbol{s}_{ij}$ is included, and its Shapley value is computed within the set $\boldsymbol{n}' = \boldsymbol{n}\backslash\{i, j\} \cup \boldsymbol{s}_{ij}$. The interaction between players i and j can be quantified by the additional reward gained when they form a coalition. The calculation formula is as follows:

$$\begin{aligned} B(\boldsymbol{s}_{ij}) &= \Phi(\boldsymbol{s}_{ij}|\boldsymbol{n}') - [\Phi(i|\boldsymbol{n}_i) + \Phi(j|\boldsymbol{n}_j)] \\ &= \sum_{s \subseteq \boldsymbol{n}\backslash\{i,j\}} \binom{n-1}{|\boldsymbol{s}|}^{-1} [\Delta f(\boldsymbol{s}, i, j)] \end{aligned} \tag{3}$$

where $\boldsymbol{n}_i = \boldsymbol{n}\backslash\{j\}$, $\boldsymbol{n}_j = \boldsymbol{n}\backslash\{i\}$, $\Delta f(\boldsymbol{s}, i, j) = f(\boldsymbol{x}_{\boldsymbol{s}\cup\{i,j\}}) - f(\boldsymbol{x}_{\boldsymbol{s}\cup\{i\}}) - f(\boldsymbol{x}_{\boldsymbol{s}\cup\{j\}}) + f(\boldsymbol{x}_{\boldsymbol{s}})$. If $B(\boldsymbol{s}_{ij}) > 0$, there is a positive game interaction between players i and j, where they cooperate to achieve higher rewards in the game. If $B(\boldsymbol{s}_{ij}) < 0$, there is a negative game interaction between players i and j, and the coalition formed by them obtains lower rewards in the game.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

TABLE I
THREE CLASSES OF POST-HOC EXPLANATION METHODS

Methods	Characteristics	Representative Methods
Back Propagation-Based	This method computes explanations at a pixel level, delivering relatively fast and accurate interpretations. It is primarily employed in the classification of neural networks. The resulting saliency maps often represent a trade-off between human interpretability and loyalty to the network's decision-making process.	Guided Backpropagation, Vanilla Gradient, and XRAI [10]
Activation-Based	The intuition behind activation-based interpretability methods is to combine only those feature maps with patterns the network deems important as output. Selecting these feature maps highlights regions in the input image relevant to the network, producing relatively easy-to-interpret heat maps that are coarse. However, they lack accuracy and are unsuitable for fine-grained explanations.	CAM series [11], [12]
Masking-Based	This model-agnostic method, relying on perturbation techniques, excels in creating more global and interpretable results independent of model type and architecture. However, compared to other methods, this method incurs a significant computational cost.	LIME [13], LRP , DeepLIFT, and SHAP [14].

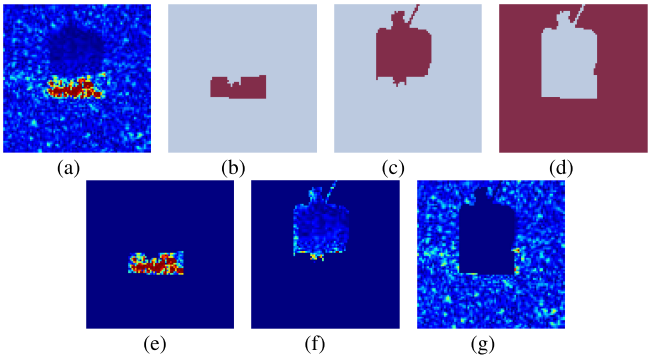


Fig. 1. Example of the SAR image decomposition. (a) $\boldsymbol{x}$. (b) $\boldsymbol{m}_{\text{target}}$. (c) $\boldsymbol{m}_{\text{shadow}}$. (d) $\boldsymbol{m}_{\text{clutter}}$. (e) $\boldsymbol{x}_{\text{target}}$. (f) $\boldsymbol{x}_{\text{shadow}}$. (g) $\boldsymbol{x}_{\text{clutter}}$.

C. Baseline Values

The Shapley value is the only attribution method that satisfies linearity, dummy, symmetry, and efficiency axioms [16], collectively considered as a fair attribution to input feature units. However, when interpreting DNN, a specific challenge arises in faithfully defining the absence of input variables. The most classical approach involves the use of **baseline values** (also called reference values) $\boldsymbol{b} = \{b_1, b_2, \ldots, b_n\}$ to denote the presence or absence of variables [17].

The objective is to learn the optimal baseline value, which can faithfully represent the absence state of input variables.

When computing the shapely value of $f(\boldsymbol{s})$, it is necessary to replace all the input feature units in $\boldsymbol{n}\backslash\boldsymbol{s}$ with the baseline values. This represents the masking effect of these input feature units while retaining the original input values on the input feature units in $\boldsymbol{s}$. Specifically, for an SAR image $\boldsymbol{x}$, $\boldsymbol{x}_{\boldsymbol{s}}$ is defined as a set of masked feature units. The output of the network in this scenario is denoted as

$$f(\boldsymbol{s}) = \begin{cases} x_i, & i \in \boldsymbol{s} \\ b_i, & i \notin \boldsymbol{s}. \end{cases} \tag{4}$$

SAR images are typically large in size and cover extensive areas. They are commonly processed into smaller "chips" for analysis, enabling the identification of components such as targets, shadows, and clutter. To analyze the contribution of different parts of SAR images to the network's decision, the set $\boldsymbol{s}$ is considered as a collection of pixels, referred to as superpixels [18]. In order to represent the target and shadow regions, object segmentation techniques are introduced to obtain mutually exclusive binary masks for the target and shadow regions, denoted as $\boldsymbol{m}_{\text{target}}$ and $\boldsymbol{m}_{\text{shadow}}$.

$$\begin{aligned} \boldsymbol{x} = &\underbrace{\boldsymbol{x} \odot \boldsymbol{m}_{\text{target}}}_{\boldsymbol{x}_{\text{target}}} + \underbrace{\boldsymbol{x} \odot \boldsymbol{m}_{\text{shadow}}}_{\boldsymbol{x}_{\text{shadow}}} \\ &+ \underbrace{\boldsymbol{x} \odot (1 - \boldsymbol{m}_{\text{target}} - \boldsymbol{m}_{\text{shadow}})}_{\boldsymbol{x}_{\text{clutter}}} \end{aligned} \tag{5}$$

where $\odot$ is the Hadamard product. Fig. 1 illustrates an instance of the decomposition. The SAR image in Fig. 1 is taken from the MSTAR dataset, and the masks are provided by the publicly accessible MSTAR segmentation annotations, specifically the SARBake dataset [19].

After SAR image decomposition, a superpixel set $\boldsymbol{n} = \{\boldsymbol{x}_{\text{target}}, \boldsymbol{x}_{\text{shadow}}, \boldsymbol{x}_{\text{clutter}}\}$ is established, where $|\boldsymbol{n}| = 3$.

Due to the unique imaging mechanism of SAR, which reflects the electromagnetic (EM) properties of the target surface, SAR images inherently lack comprehensive geometric structures and color information of the targets. As discussed before, SAR image can be simplified into three distinct regions: target, shadow, and clutter. The target region provides information about the shape, size, and structure of the target, with different target types exhibiting unique scattering patterns in the image, primarily used for target recognition. The shadow region is caused by variations in target height or terrain, indirectly providing information about the target's shape, size, and three-dimensional geometry, and directly supplying information about the surface height and terrain undulations. The clutter region mainly includes scattering noise caused by natural features on the Earth's surface, such as vegetation and land cover. This clutter may exhibit complex textures in the image due to the unevenness of surface coverage. Background clutter may also encompass signal interference caused by atmospheric disturbances, instrument artifacts, and other sources.

Baseline values such as zero values and mean values are commonly used to mask input variables. However, the traditional masking methods always fail to faithfully represent states of information absence. The comparison of causal

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

information in the masked samples with that in the original samples is conducted to assess the following baseline values.

1) Zero Baseline Values: The pixel values in the masked region are set to zero

$$b_i = 0 \quad \forall i \notin \boldsymbol{s}. \tag{6}$$

Zero baseline may introduce additional signals to the input image, such as black spots.

2) Mean Baseline Values: The pixel values in the masked region are set to the mean value of the corresponding region pixel values across all variables. For instance, the mean of the target region pixel values across all samples is used as the baseline value for the target region

$$b_i = \mathbb{E}_{\boldsymbol{X}}[x_i] \quad \forall x_i \in \boldsymbol{X} \tag{7}$$

where $\boldsymbol{X}$ represents a collection of images. However, this method, similar to the zero baseline, introduces additional signals to the input. For example, the mean value may introduce significant gray spots in the image and potentially form new edges as anomalous causal patterns.

3) Random Baseline Values: Utilizing a certain random distribution to generate baseline values and subsequently replacing the pixel values in the masked region

$$b_i = p, p \sim P \tag{8}$$

where P represents a certain distribution. For instance, adopting a Gaussian distribution as the masking baseline. However, this method removes high-frequency information while preserving low-frequency information.

Based on the above analysis, there is often an overlooked issue with masking-based methods. The credibility of attributions heavily relies on whether the current baseline values truly eliminate the signal functionality of input variables. This must occur without introducing out-of-distribution (OOD) scenarios. Currently, there is no theoretical framework to verify whether baseline values faithfully represent the absence of input variable signals, that is, the removal of all signals from the input variables.

Most studies [20], [21] have compared the impact of different baselines on explainability but have not provided principled guidelines for setting baseline values. Shrikumar et al. [7] introduced DeepLIFT to estimate attributions of input variables and discussed the choice of baseline values. Additionally, Agarwal and Nguyen [22] and Frye et al. [23] addressed OOD issues caused by baseline values using generative models. Ren et al. [17] rethought and formulated baseline values from a game-theoretic causal perspective. They defined a missing state for input variables and proposed a method based on causal pattern quantity learning for optimal baseline value determination.

Different from prior research predominantly conducted in the optical image domain, the use of SHAP methods for image masking in SAR images poses a more pronounced OOD challenge. This is due to the unique background clutter distribution inherent in SAR image. Drawing inspiration from Ren et al. [17], the reengineering of baseline values is approached from the perspectives of multivariate interactions and clutter distribution. Subsequently, expedited baseline learning based on causal effect scores is achieved by reducing the quantity of causal patterns (number of superpixels).

III. Methodology

In this section, the overall framework and process of the proposed method are first presented. Subsequently, the key algorithmic components of the framework are described.

A. Overview of Proposed Method

As illustrated in Fig. 2, the proposed general workflow relies on four key steps.

In the first step, the clutter modeling of the SAR dataset is undertaken to mitigate the discrepancy between the background clutter distribution and the distribution of baseline values. A detailed description of this step is provided in Section III-B4.

The second step involves the generation of baseline values, which is the primary innovation of this article. This step entails creating masked samples of the original image. The mask consists of superpixels, with their shape and quantity determined by a superpixel mask matrix. The extent of obscuration is dictated by the baseline value matrix, which matches the dimensions of the original image. Only when the areas specified by the superpixel mask matrix necessitate masking will the corresponding elements from the baseline value matrix be applied to mask the original image. The module responsible for generating masked samples outputs a series of samples used to compute interaction scores among individual superpixels within the samples. These scores are jointly determined by a multivariate interaction loss and the maximum mean discrepancy (MMD) [24], [25] loss between clutter distribution and baseline value distribution. Smaller values of the interaction scores indicate less information introduced by the baseline values, making them more suitable as proposed masking strategies. Superpixels in masked samples corresponding to lower interaction scores are treated as coalitions. This reduces the number of possible superpixel combinations and thereby decreases the sample count, contributing to algorithm acceleration. Subsequently, the superpixel mask matrix is updated iteratively.

In the third step, it is required to specify a region in the image, as the region of interest (ROI). Combining this ROI with the baseline value matrix yields a masked image for the input picture.

In the fourth step, the SHAP algorithm is employed to calculate Shapley values, revealing the contribution of the ROI in the input image to model decisions. This comprehensive four-step process aims to enhance the explainability and efficiency of the algorithm in analyzing SAR datasets.

Moreover, the region selection step differs from cases of image segmentation. It involves a complete partitioning of the image during a single explanation process. This means the partitioned regions must not have any intersections. This method is recommended primarily for image classification. It aims to elucidate the decision-making process of the entire model.

The masked sample generation module involves overlaying the original image with a set of uniformly distributed regular

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

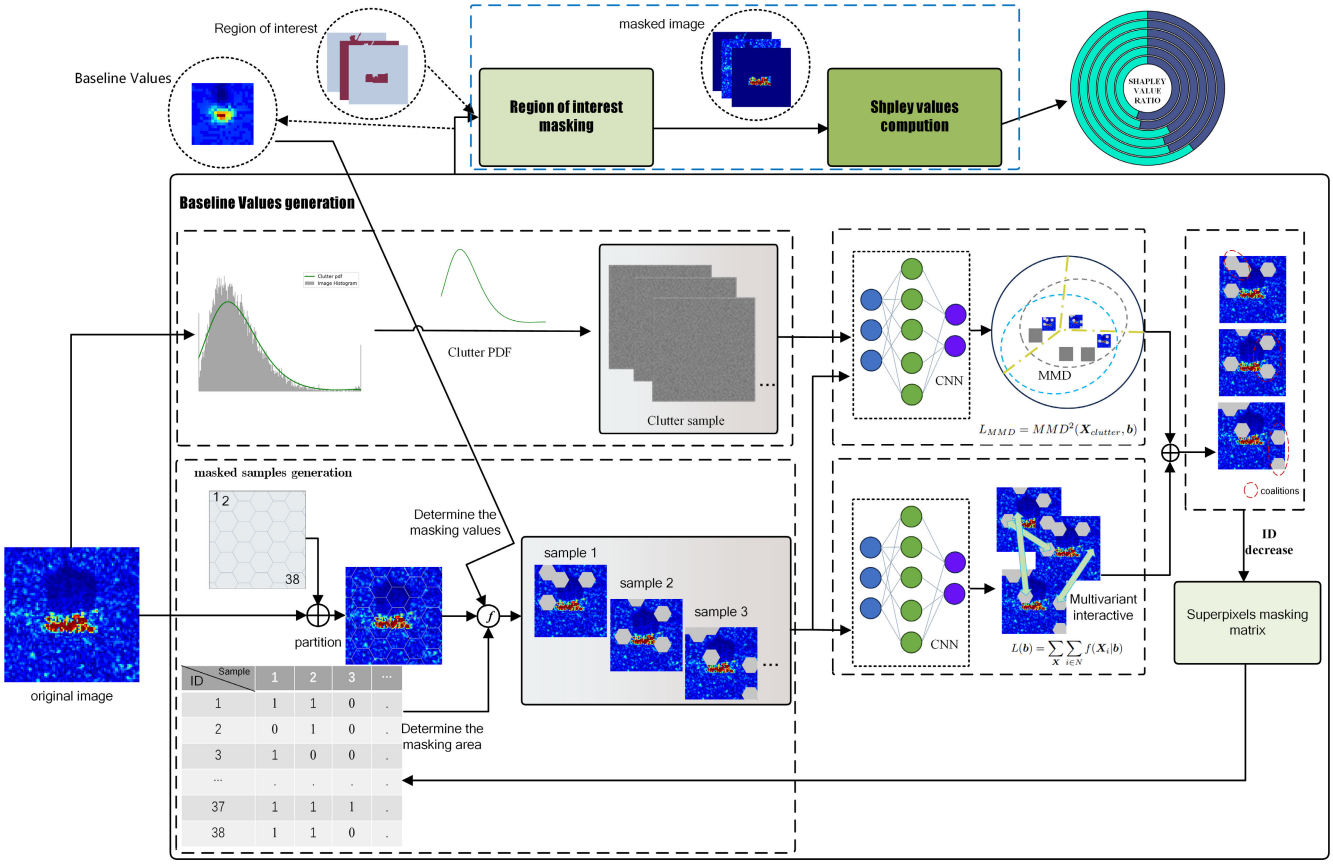


Fig. 2. Proposed general workflow.

hexagons, treating each pixel within a hexagon as a superpixel. Each superpixel is segmented according to a hexagonal grid. The masked samples are then generated through the function f, utilizing a masking matrix to determine the obscured regions and employing the baseline value matrix as the numerical values for obscuration.

In the discussion concerning the shape of superpixels, inspiration was drawn from literature [26]. Since SHAP values depend on the surface of superpixels, this implies that a more uniform shape of superpixels would facilitate human interpretation. Hexagonal grids are deemed more suitable for visual interpretation, primarily due to their connectivity patterns.

Furthermore, our method can also be applied to improve DeepLIFT by utilizing the computed baseline values. DeepLIFT is a local explanation method that relies on backpropagation. We suggest replacing the baselines in the DeepLIFT algorithm with the baseline values obtained from the SHAP method, as depicted in Fig. 2 (Replace the method inside the blue dotted line with the DeepLIFT method). This substitution enables more accurate and reliable explanations.

B. Causality-Erasing SHAP

This section sequentially expounds on the modeling and learning of baseline values, efficiency optimization in baseline value computation, and the constraint of baseline values through clutter distribution.

1) Decompose the Output Into Sparse Interactions: Ren et al. [27] have demonstrated that given a trained DNN f and an input $\boldsymbol{x}$ with $|\boldsymbol{n}|$ input variables, the DNN output $f(\boldsymbol{x})$ can be decomposed into the influences of interactions among the input variables. For example, suppose the SAR images of the MSTAR dataset are decomposed into targets, shadows, and background clutter, that is, $\boldsymbol{s} = \{\boldsymbol{x}_{\text{targets}}, \boldsymbol{x}_{\text{shadows}}, \boldsymbol{x}_{\text{clutter}}\}$. Only when the input superpixel set contains all three elements of $\boldsymbol{s}$ appearing simultaneously, the causal pattern $\boldsymbol{s}$ will be triggered. It exerts an influence $U_{\boldsymbol{s}}$ on the final classification accuracy of the network. Otherwise, the absence of any element in the set $\boldsymbol{s}$ will remove $U_{\boldsymbol{s}}$ and generate a new causal effect $U'_{\boldsymbol{s}}$. This may significantly reduce the corresponding final classification accuracy.

Specifically, let $\boldsymbol{s} \subseteq \boldsymbol{n}$ represents a subset of input variables. The interactions among variables in $\boldsymbol{s}$ are defined as the Harsanyi dividend [28].

$$U_{\boldsymbol{s}} = \sum_{\boldsymbol{s}' \subseteq \boldsymbol{s}} (-1)^{|\boldsymbol{s}|-|\boldsymbol{s}'|} f(\boldsymbol{x}_{\boldsymbol{s}'}). \tag{9}$$

According to the definition provided above, the following equation can be obtained:

$$f(\boldsymbol{x}) = \sum_{\boldsymbol{s} \subseteq \boldsymbol{n}} U_{\boldsymbol{s}}. \tag{10}$$

The following proves that masking input hyperparameters eliminates causal effects.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

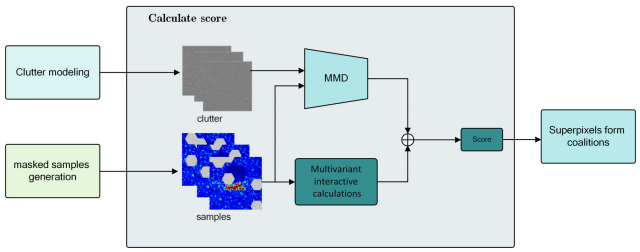


Fig. 3. Flowchart for calculating score.

Let $\boldsymbol{s} = \boldsymbol{s}' \cup \{i\}$, if $i \in \boldsymbol{s}$ is masked, then $\forall \boldsymbol{l}$ s.t. $i \notin \boldsymbol{l}$, $\boldsymbol{x}_{\boldsymbol{l}} = \boldsymbol{x}_{\boldsymbol{l} \cup \{i\}}$. Therefore, $f(\boldsymbol{l} \cup \{i\}) = f(\boldsymbol{l})$. According to (9), the following equation can be obtained:

$$\begin{aligned} U_{\boldsymbol{s}} &= \sum_{\boldsymbol{l} \subseteq \boldsymbol{s}} (-1)^{|\boldsymbol{s}|-|\boldsymbol{l}|} f(\boldsymbol{l}) \\ &= \sum_{\boldsymbol{l} \subseteq (\boldsymbol{s}' \cup \{i\})} (-1)^{|\boldsymbol{s}'|+1-|\boldsymbol{l}|} f(\boldsymbol{l}) \\ &= \sum_{\boldsymbol{l} \subseteq \boldsymbol{s}'} (-1)^{|\boldsymbol{s}'|+1-|\boldsymbol{l}|} f(\boldsymbol{l}) + \sum_{\boldsymbol{l} \subseteq \boldsymbol{s}'} (-1)^{|\boldsymbol{s}'|-|\boldsymbol{l}|} f(\boldsymbol{l}) \\ &= \sum_{\boldsymbol{l} \subseteq \boldsymbol{s}'} \left((-1+1)(-1)^{|\boldsymbol{s}'|-|\boldsymbol{l}|} \right) f(\boldsymbol{l}) \\ &= 0 \end{aligned}$$

2) Learning the Baseline Values: In the context of classical game theory scenarios, the introduction of Shapley values does not necessitate the definition of player absence. According to [17], the true masking state of features should yield the most simplified causal explanations. Correct baseline values typically generate the simplest causal explanations. This means using the least number of causal patterns to explain DNN behavior. In contrast, if incorrect baseline values fail to completely eliminate all effects of the AND relationships of features, the residual effects will be erroneously interpreted as numerous other redundant patterns.

Based on the above analysis, it can be summarized that simpler causal relationships with the minimum causal patterns are more likely to represent the essence of DNN inference logic. This also enables us to consider baseline values that minimize the number of significant causal patterns (i.e., achieve the simplest causal relationships) as the optimal baseline values.

Therefore, learning baseline values can be formulated as sparsing causal patterns in deep models. It is noteworthy that baseline values obtained in this manner, when masking causal effects, do not introduce excessive new effects

$$\boldsymbol{b}^* = \arg\min_{\boldsymbol{b}} \sum_{\boldsymbol{X}} |\Omega(\boldsymbol{X})| \tag{11}$$

$$\Omega(\boldsymbol{X}) = \{\boldsymbol{s} \subseteq \boldsymbol{n} || U(\boldsymbol{X}; \boldsymbol{b})| > \tau\} \tag{12}$$

where $\boldsymbol{b}^*$ represents the optimal baseline values, $|\Omega(\boldsymbol{X})|$ denotes the number of causal patterns, and τ is the threshold for feature interactions.

3) Estimating Baseline Values: Learning the optimal baseline value based on (11) and (12) incurs an exponential computational cost when enumerating all causal patterns. Therefore, approximate solutions for learning baseline values are needed.

If m' incorrect baseline values are used to calculate causal effects, that is, the equation $f(\boldsymbol{x}_{\boldsymbol{s}}) = w_{\boldsymbol{s}} \Pi_{j \in \boldsymbol{s}} (\boldsymbol{x}_j - \delta_j)$, there are a total of $\binom{m'}{k - |\boldsymbol{s}| + m'}$ occurrences of kth-order causal patterns, where $k > |\boldsymbol{s}| - m'$. A kth-order causal pattern implies that this pattern represents an AND relationship among k superpixels [17].

According to above, incorrect baseline values often erroneously interpret higher-order causal patterns as an unnecessary abundance of lower-order patterns, where the order m of the causal effect $U_{\boldsymbol{s}}$ is defined as the cardinality of $\boldsymbol{s}$, that is, $m = |\boldsymbol{s}|$.

Therefore, the objective of learning baseline values is roughly equivalent to penalizing the influence of lower-order causal patterns. This helps prevent the learning of incorrect baseline values that mistakenly represent higher-order patterns as an exponential quantity of lower-order patterns

$$\min L(\boldsymbol{b}) \tag{13}$$

$$L(\boldsymbol{b}) = \sum_{\boldsymbol{X}} \sum_{\boldsymbol{l} \subseteq \boldsymbol{s}} |U(\boldsymbol{X}|\boldsymbol{b})|. \tag{14}$$

Optimizing the above equation directly is NP-hard as each input SAR image consists of a large number of superpixel groups. To address this challenge, SAR images were decomposed into sets of superpixels. Each superpixel set is considered a group of causal patterns. This method aligns with the goal of exploring the contributions from different regions. Interestingly, when $\boldsymbol{s}' = \boldsymbol{s}$, (9) can be simplified as

$$\begin{aligned} U_{\boldsymbol{s}} &= f(\boldsymbol{x}_{\boldsymbol{s}}|\boldsymbol{b}) - f(\phi) \\ &= \Delta f(\boldsymbol{x}_{\boldsymbol{s}}|\boldsymbol{b}) \end{aligned} \tag{15}$$

Substituting (15) into (14) results as

$$L(\boldsymbol{b}) = \sum_{\boldsymbol{X}} \sum_{i \in N} f(\boldsymbol{X}_i|\boldsymbol{b}) \tag{16}$$

where $\boldsymbol{X}$ represents the set of images, and N is the number of superpixel sets.

4) Baseline Values Correction: The background clutter in SAR images is widely present [8], whether in the background, target, or shadow regions. This clutter characteristic is unique to SAR images and is not present in optical imagery. Therefore, when performing masking operations on SAR images, the influence of clutter on the distribution of the masked SAR image needs to be considered. If the baseline value does not account for the clutter distribution, the clutter distribution in the masked regions may disappear, and the network may focus on the vanished clutter distribution. This introduces new non-causal information to the network, contradicting the principle of introducing zero information through baseline values. To ensure that the baseline value carries the background clutter distribution information of SAR images, it is necessary to reduce the difference between the background clutter distribution and the baseline value distribution.

In order to enhance states with no informative content, an attempt is made to filter out causal influences from the features in the latent space. Target samples aligned at the class level in the output space are identified based on these uninformative states, safeguarding their semantic consistency against

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON ABOUT PREVIOUS MASKING METHODS

Setting of Baseline values	Baseline values are constant or not	Different samples share the Baseline value or not	Shortcomings
Zero Baseline values	✔	✔	introduce additional signals
Mean Baseline values	✔	✔	introduce additional signals
Random Baseline values	✔	✗	not remove low-frequency components
Proposed method	✔	✔	high computational cost

TABLE III
DECREASE OF RECOGNITION ACCURACY (%) UNDER CLUTTER REPLACEMENT OF MODELS

Model	Baseline Acc.	Params	Baseline value			
			Zero	Mean	Random	**Proposed**
A-ConvNet	95.60	0.30M	63.04 ↓	66.21 ↓	42.32 ↓	**25.30** ↓
AM-CNN	96.41	2.60M	69.33 ↓	68.81 ↓	46.61 ↓	**28.21** ↓
EfficientNet-B0	92.54	4.02M	70.24 ↓	67.72 ↓	41.73 ↓	**28.39** ↓
ResNet18	95.01	12.43M	74.24 ↓	70.42 ↓	52.28 ↓	**32.66** ↓
MVGGNet	95.10	16.81M	74.00 ↓	70.23 ↓	53.50 ↓	**29.79** ↓
ConvNeXtTiny	97.16	27.83M	73.63 ↓	72.19 ↓	51.14 ↓	**34.39** ↓

the side effects of background clutter distribution. In essence, the purpose of using MMD is to approximate the distribution of baseline values to follow the background clutter distribution, distancing itself from non-background clutter distributions.

The clutter distribution is first modeled to obtain an approximate distribution, such as Rayleigh distribution, Weibull distribution, K-distribution [29], etc. Then, based on the prior distribution, maximum likelihood estimation is employed to obtain parameter estimates of the distribution. Thus, the clutter distribution statistical function for SAR image data is derived.

Subsequently, the loss function of (16) is modified as

$$L(\boldsymbol{b}) = \sum_{\boldsymbol{X}} \sum_{i \in N} f(\boldsymbol{X}_i|\boldsymbol{b}) + \lambda L_{\text{MMD}} \tag{17}$$

$$L_{\text{MMD}} = \text{MMD}^2(\boldsymbol{X}_{\text{clutter}}, \boldsymbol{b}) \tag{18}$$

where λ is the weight parameter, used to measure the importance of both the marginal contribution and the superpixel distribution. In experiments, λ is set to 0.2. MMD is a test statistic to decide the similarity of two distributions.

In (17) and (18), $\boldsymbol{X}_{\text{clutter}}$ represents the dataset of background clutter, and $\boldsymbol{b}$ denotes the region of the generated sample subject to masking. The values within this region are filled with baseline values generated in the previous iteration. As shown in Fig. 3, the operation of MMD imposes the background clutter distribution information to the distribution of baseline values. If the generated baseline values faithfully reflect the information-absent state, the contribution $\Phi(\boldsymbol{b}|\boldsymbol{n})$ for the occluded region of the corresponding sample is 0. This often results in lower interaction scores. Exploiting the characteristics of clutter distribution in SAR images involves the elimination of causal patterns from baseline values, thereby obtaining a specific masking method tailored for SAR images. The overall algorithmic workflow is illustrated in Algorithm 1.

5) Comparison and Analysis: Table II summarizes the comparison between the proposed masking method and four previous masking methods. It is evident that the proposed method, without introducing additional noise, maintains constant baseline values that are shared across different samples.

Table III presents the variation in accuracy across different network architectures. The baseline accuracy represents the predictive accuracy on the original test set using the original training set. The subsequent columns are as follows: when the network parameters are frozen, the test set's clutter regions are masked under various baseline values masking methods. It is observed that accuracy decreases in all masking methods, but proposed method consistently exhibits the smallest decline across all network architectures. Therefore, two conclusions can be drawn: 1) in the MSTAR dataset, clutter regions play a crucial role in the network's decision-making process, indicating a potential overfitting of the network to clutter; and 2) proposed masking method introduces less non-causal information, enabling the network to rely more on the target, resulting in the minimal decline in accuracy.

Algorithm 1 Causality-Erasing SHAP Flow

Input: $\boldsymbol{X}$: Origin images; $\boldsymbol{Y}$: Lables; $\boldsymbol{X}_{s \subseteq n}$ Masked images; $\boldsymbol{X}_{clutter}$: Clutter; lr: Learn rating; Itr: Iteration ordinal number; λ; f: Trained model;

Output: The Shapley value $\boldsymbol{\Phi}(i|\boldsymbol{n})$ of each superpixel feature;

The Baseline values are initialized with the dimensions of the image, and the initial elements are set to either the Mean or Zero values.
for $itr < Itr$ **do**
  $\Delta f(\boldsymbol{X}_{i|s}) \leftarrow f(\boldsymbol{X}_{s\cup\{i\}}) - f(\boldsymbol{X}_s)$
  $\boldsymbol{\Phi}(i|\boldsymbol{n}) \leftarrow \sum_{s \subseteq n \setminus \{i\}} \binom{n-1}{|s|}^{-1} \Delta f(\boldsymbol{X}_{i|s})$
  $MMD^2(\boldsymbol{X}_{clutter}, \boldsymbol{b}) \leftarrow \|\mathbb{E}_{\boldsymbol{X}_{clutter}}[\phi(X_{clutter}) - \phi(X^{\boldsymbol{b}})]\|^2_{\mathcal{H}}$
  $MMD^2(\boldsymbol{X}_j, \boldsymbol{b}) \leftarrow \|\frac{1}{|s|}\sum_{j=1}^{|s|}\phi(X_j) - \phi(X^{\boldsymbol{b}})\|^2_{\mathcal{H}}$
  $L_{MMD} \leftarrow MMD^2(\boldsymbol{X}_{clutter}, \boldsymbol{b})$
  $L(\boldsymbol{b}) \leftarrow \sum_{\boldsymbol{X}} \sum_{i \in N} f(\boldsymbol{X}_i|\boldsymbol{b}) + \lambda L_{MMD}$
  grads $\leftarrow L(\boldsymbol{b})$
  $\boldsymbol{b} \leftarrow \boldsymbol{b} - lr * gards$
  itr $\leftarrow$ itr $+1$
end for
get $\boldsymbol{b}^*$
calculate $\boldsymbol{\Phi}(i|\boldsymbol{n})$

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section first introduced the two datasets, along with their respective data partitioning schemes and then setup of the experimental environment. Following this, an assessment for the loyalty of the baseline values is carried out. Then, explainability analysis is conducted on the two datasets. Lastly, the reliability of the proposed algorithm is further validated by comparing the average recognition accuracy under different baseline values.

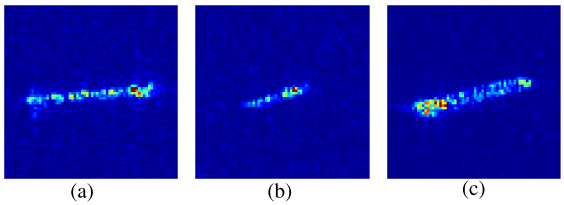
A. Datasets and Setting

The experiment will utilize the MSTAR dataset. The MSTAR dataset has a resolution of 0.3 × 0.3 m and employs the HH polarization mode. This dataset is employed for salient region experiments and all quantitative trials. Under standard operating conditions (SOCs), the MSTAR dataset

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
DESCRIPTION TRAINING AND TEST SET

Type	Class	Training set(17°)	Test set(15°)
Rocket launcher	2S1	299	274
Armored personnel carrier	BMP2	698	587
	BRDM2	298	274
	BTR70	233	196
	BTR60	256	195
Bulldozer	D7	299	274
Tank	T72	691	582
	T62	299	273
Truck	ZIL131	299	274
Air defense unit	ZSU234	299	274
	Total	3671	3203



(a) (b) (c)

Fig. 4. Three classes of targets in the OpenSARShip dataset. (a) BulkCarrier. (b) Cargo. (c) Container.

TABLE V
DESCRIPTION OF THE TRAINING AND TEST SET

Class	Training set	Test set
BulkCarrier	160	40
Cargo	160	40
Container	160	40
Total	480	120

encompasses ten target types under various scenes, observation conditions, rotation angles, and elevation angles. It contains radar echo raw data and SAR images generated through image processing.

The dataset is divided into training and testing sets, with slight variations in the quantity of classes between the two sets. Table IV presents the distribution details of various target classes within the training and testing sets of the MSTAR dataset. Amoon and Rezai-Rad [30] provide a solution for segmenting the MSTAR dataset. However, compared to the SARBake segmentation, it is not detailed enough because the SARBake solution uses a real mask matrix for segmentation, which theoretically can be taken as the ground truth for segmentation.

To further demonstrate the effectiveness of the recommended baseline values, a series of experiments were conducted on the OpenSARShip dataset generated by Shanghai Jiao Tong University in 2017 [31].

OpenSARShip-1.0 dataset comprises 11 346 image slices of 17 ship classes (with one class representing all unknown ships) possessing VH and VV polarizations from the Sentinel-1 satellite. In comparison to the MSTAR dataset, this dataset exhibits a more complex background, and more variability in shapes within the same class. For the segmentation of the OpenSARShip dataset, we used SISNet [32]. Since the OpenSARShip dataset ignores the influence of the shadow area cast by the ship at sea, the dataset is only divided into target region and clutter region.

Given the uneven distribution of samples across different classes, three classes (BulkCarrier, Cargo, Container) are selected as they have the largest sample size. These samples were then randomly split in an 8:2 ratio for training and testing sets, respectively. Fig. 4 illustrates sample images for the three classes, while Table V provides detailed information on their partitioning. Due to significant variations in image sizes within the OpenSARShip dataset, images were resized to 128 × 128 by cropping or padding them.

All experiments were conducted on the server with the following configuration: NVIDIA GeForce RTX 3090 GPU × 2, an AMD Ryzen 5900X CPU, and the Ubuntu 18.04.5 LTS operating system. The coding implementation utilized the PyTorch framework.

B. Validate the Loyalty of Baseline Values

1) Qualitative Evaluation on Different Masking Methods: To validate the accuracy of the proposed baseline values, three explainability methods, namely Vanilla Gradient, Guided Backpropagation, and XRAI [10], were employed. Visualization was performed on the same SAR image (BTR60-HB03798.003, with the clutter region masked). The results are illustrated in Fig. 5. The baseline network is denoted as A-ConvNet [33]. All three mentioned methods are capable of generating saliency maps, according to which we can analyze the regions of input features that the network focuses on.

Each column of Fig. 5 presents visualizations for the original image and different masking methods applied to the clutter region, namely zero, mean, random, and the proposed, respectively. Both Vanilla Gradient and Guided Backprop methods primarily focus on the target region in the images. The Vanilla Gradient method tends to concentrate on finer-grained target features, appearing more dispersed. In contrast, the Guided Backprop method is more adept at revealing the complete outline of the target. However, upon observing the clutter region, the previous three masking methods exhibit scattered noise attention.

The zero and mean masking methods display a wide-ranging distribution of noise, introducing the additional noise information to the images. While the random masking method eliminates high-frequency noise, there still remain some low-frequency noise. For the XRAI method, the salient regions (highlighted in orange in the figure) for the first three masking methods show a shift toward the clutter area. They fail to cover the entire target region, indicating an attention imbalance caused by the introduction of extra information. In contrast, the proposed masking method shows saliency maps that solely focus on the non-masking areas. This method can accurately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

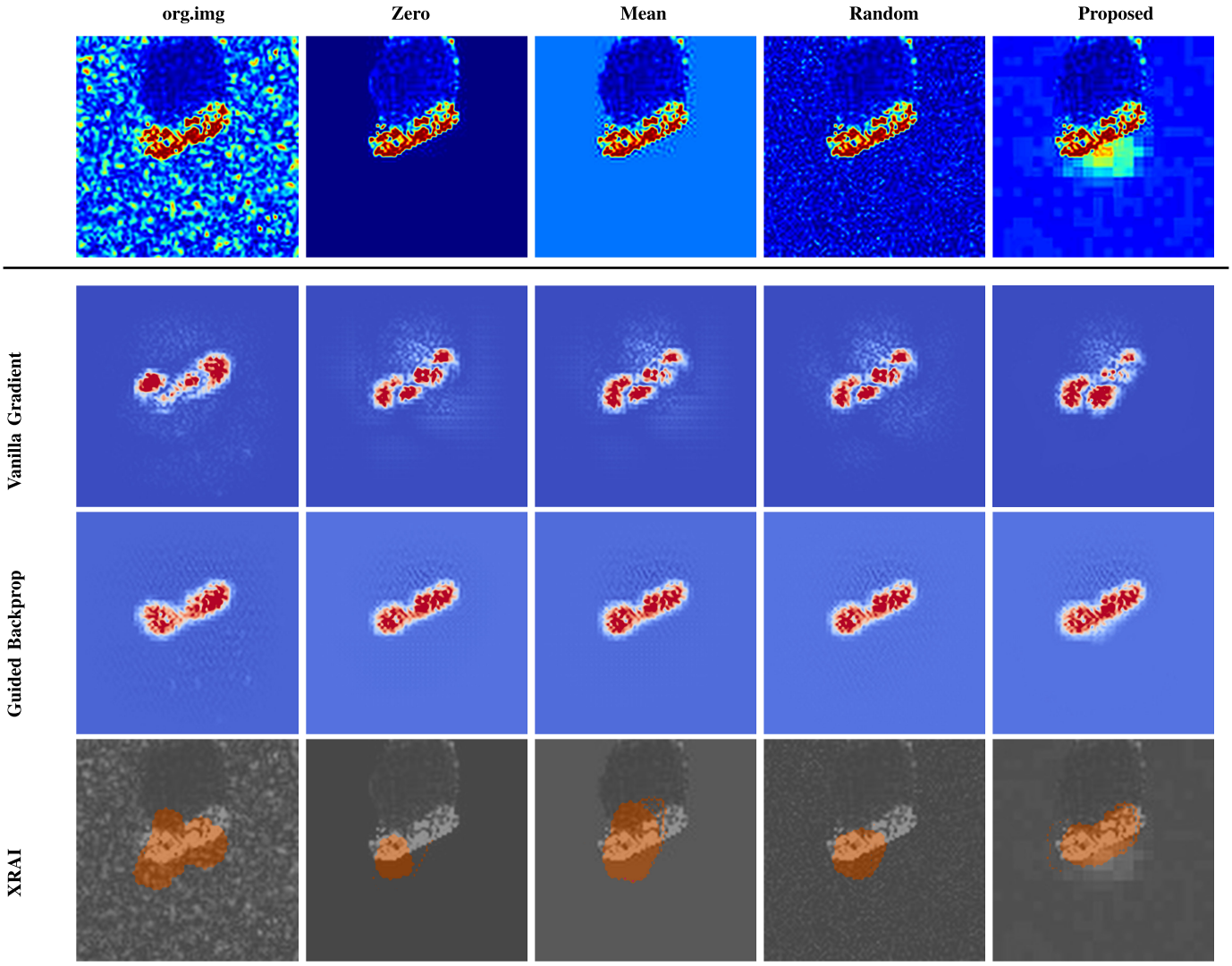


Fig. 5. Vanilla Gradient Guided Backprop and XRAI results via A-ConvNet.

delineate the target region, suggesting that it does not introduce additional information. As a result, it provides a more faithful baseline for attributing contributions using the SHAP method in subsequent analyses.

2) Quantitative Evaluation on Different Masking Methods: To assess whether the above baseline values eliminate all causal patterns in the original input and whether they introduce new causal patterns, the following criteria are employed to evaluate the quality of the baseline values.

$$R = \mathbb{E}_x \left[\frac{\sum_{\boldsymbol{s} \subseteq \boldsymbol{n}} |U'_{\boldsymbol{s}}| - \sum_{\boldsymbol{s} \subseteq \boldsymbol{n}} |U_{\boldsymbol{s}}^{(\text{noise})}|}{\sum_{\boldsymbol{s} \subseteq \boldsymbol{n}} |U_{\boldsymbol{s}}|} \right]. \tag{19}$$

A set of samples were generated based on the original image $\boldsymbol{x}$, where specific regions were masked. $U'_{\boldsymbol{s}}$ denotes the causal effects in such masked samples. $U_{\boldsymbol{s}}$ represents the causal effects in the original sample $\boldsymbol{x}$, used for normalization. $U\boldsymbol{s}^{(\text{noise})}$ represents the causal effects in the white noise input, indicating the inevitable influence of a substantial amount of noise patterns. Therefore, the $U_{\boldsymbol{s}}^{(\text{noise})}$ term is considered an unavoidable anchor and is removed from R for a more compelling evaluation.

TABLE VI

RATIO R BETWEEN THE REMAINING CAUSAL EFFECTS AND THE NEWLY INTRODUCED CAUSAL EFFECTS IN THE MASKED INPUT

	R_{zero}	R_{mean}	R_{random}	$R_{proposed}$
MSTAR	1.61	1.58	0.93	**0.52**
OpenSARShip-1.0	1.42	1.33	1.02	**0.61**

The masking method has two impacts on causal patterns: 1) eliminating all existing significant patterns in the original sample and 2) do not expect the masking method to introduce new significant patterns.

Thirty images were used for calculation of R, involving the random selection of three images from each class in the MSTAR dataset. The MSTAR dataset was initially divided into three parts: targets, shadows, and clutter. For each image, the clutter region was masked using zero baseline, mean baseline, random baseline, and proposed baseline, respectively. Subsequently, R_{zero}, R_{mean}, R_{random}, and R_{proposed} were calculated using (19).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

Table VI indicates that the ratios R using previous baseline values are significantly higher. Although the masking method based on the random distribution performs better than some other baselines, the proposed method exhibits the optimal performance. It suggests that previous masking methods either failed to mask existing patterns or triggered new patterns.

C. Causal Effect Exploration of MSTAR Dataset

In the testing set of the MSTAR dataset, the images were initially segmented into three regions: target, shadow, and clutter, and the masking method is applied to every region. Subsequently, the original training set is input into the baseline network model separately. Training is conducted without utilizing data augmentation techniques, and the network parameters are frozen. Following this, Shapley values are computed for the three regions individually on the testing set, as presented in the "Region" column of Table VII.

Next, maintaining the same experimental configuration, the testing set is partitioned into three coalition combinations: clutter-target, target-shadow, and shadow-clutter. Utilizing (3), their binary Shapley values are computed, as displayed in the "Coalition" column of Table VII.

The baseline models include three deep learning models from SAR-ATR, namely A-ConvNet, AM-CNN, and MVGGNet. Additionally, other models from computer vision, including EfficientNet, ResNet, and ConvNeXtTiny [34], are employed for comparative analysis.

Based on Table VII, disregarding the inter-region associations, the impact of the shadow region on recognition is minimal, a finding akin to prior research [8]. Across the A-ConvNet, AM-CNN, and EfficientNet-B0 models, it can be observed that the target region predominantly contributes to the output. Conversely, for the ResNet18, MVGGNet, and ConvNeXtTiny models, the clutter region plays a primary role in influencing the output. Table III shows that the model parameter count increases sequentially from top to bottom, which suggests that the size of network parameters influences the ROI in the images. As the model parameter count grows, the fitting capability to the clutter region strengthens. On average across the ten classes, the model relies more on the clutter region to make decisions. For the MVGGNet and ConvNeXtTiny models, compared to the less parameterized ResNet18, there is a greater emphasis on the target region.

It may be because the unique pre-training mechanisms of MVGGNet and ConvNeXtTiny models constrain the models to focus more on target regions with texture and shape attributes. MVGGNet utilized optical pre-training weights in its original paper, and optical pre-training weights were also employed in the training of ConvNeXtTiny. Li et al. [35] demonstrate that convolutional kernels with pre-training exhibit superior texture and shape attributes compared to direct training on the MSTAR dataset. Therefore, it can be further concluded that optical pre-training weights reduce overfitting to clutter.

As indicated by the coalition columns, there is a more pronounced causal interaction between the target and shadow regions. However, the ResNet18 considers causal interaction between background clutter and the target more pivotal for decision-making. This aligns with intuitive expectations as shadows representing the projection of the target on the ground, containing to some extent information about the target. For instance, the shape and height of the target can be inferred from the shadow since the collected geometric configuration is known. The additional information can enhance target detection and classification efficiency.

Fig. 6 presents the Shapley values ratio for individual classes of MSTAR across different network architectures. Due to the lack of comparability between Shapley values among different classes, they were normalized into a ratio form. Negative Shapley values ratio indicates a negative impact, suggesting that replacing a region with the proposed baseline value helps in identifying this class. This finding was further validated in subsequent experiments.

Overall, the classes BTR70, BMPR2, and BRDM2 exhibit a phenomenon where clutter regions contribute more significantly than target regions. In certain networks, there is even a negative impact observed in the target regions, indicating that these regions not only fail to contribute to the output but also confuse the network's decision.

Due to the background clutter correlations present in both the training and testing sets for these classes in the MSTAR dataset, the classification performance of the model for these classes does not degrade. On the contrary, the deep network's ability to fit the background clutter regions actually maintains high classification performance. In the field of optics, a similar case is observed in the classification story of huskies and wolves [36]. DNNs possess strong fitting capabilities, but without proper constraints, this ability may lead to illogical non-causal relationships. This could pose a significant challenge for interpretability. Fortunately, in the AM-CNN network architecture, there exists a lightweight attention mechanism that captures visual structures in the image, providing constraints on the regions where the network focuses on [37].

Interestingly, there is also an explanation for this phenomenon from the perspective of SAR data itself. In most cases, both target and clutter regions contain a significant portion of the information required for classification. The fact that clutter regions contribute substantial information for CNNs might imply the utilization of multipath information, the presence of multipaths in the image background that also contain EM scattering information about target 3-D structure, such as the height of scattering bodies.

D. Causal Effect Exploration for OpenSARShip-1.0 Dataset

Similar to the experimental setup for the MSTAR dataset, the influence in the shadow regions cast by ships on the sea is disregarded for the OpenSARShip-1.0 dataset. The dataset is partitioned into target and clutter regions. Subsequently, Shapley values for each of these regions are computed separately on the test set, as illustrated in Fig. 7.

A concentric ring chart is used to represent the average contribution proportions of target and clutter regions under different network architectures. From the outermost to the innermost ring, the networks considered are A-ConvNet,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
SHAPLEY VALUE RATIO AND BIVARIATE SHAPLEY INTERACTION OF MODELS

Model	Region			Coalition		
	Clutter	Target	Shadow	Clutter&Target	Target&Shadow	Shadow&Clutter
A-ConvNet	24.94%(2.00)	**52.86%(4.24)**	22.19%(1.78)	1.96	**4.07**	1.62
AM-CNN	29.32%(2.09)	**46.59%(3.33)**	24.08%(1.72)	0.93	**1.62**	0.34
EfficientNet-B0	39.92%(5.25)	**47.31%(6.23)**	12.77%(1.68)	1.88	**2.04**	0.91
ResNet18	**49.74%(2.18)**	23.79%(1.05)	26.47%(1.16)	**1.38**	0.67	0.96
MVGGNet	**45.39%(7.30)**	39.36%(6.33)	15.25%(2.45)	0.41	**1.57**	1.00
ConvNeXtTiny	**42.54%(3.16)**	31.48%(2.34)	25.98%(1.93)	0.53	**1.52**	0.82

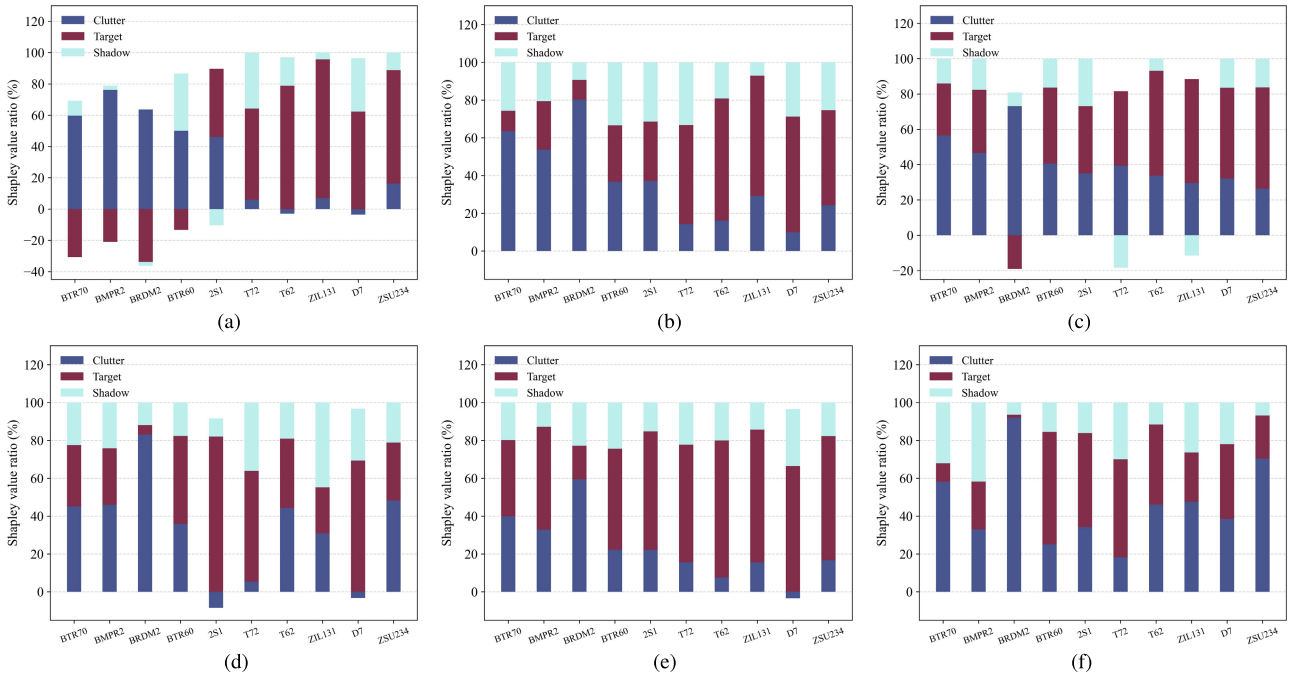


Fig. 6. Shapley value ratio of different DNNs for MSTAR. (a) A-ConvNet. (b) AM-CNN. (c) EfficientNet-B0. (d) ResNet18. (e) MVGGNet. (f) ConvNeXtTiny.

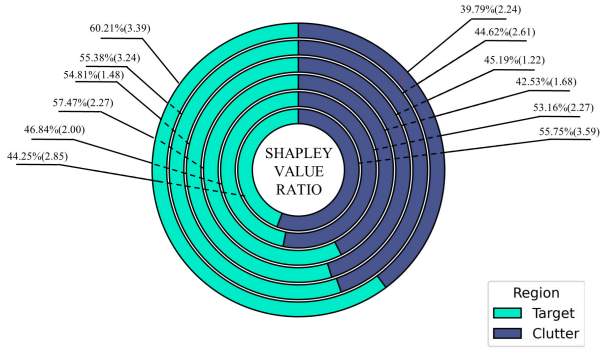


Fig. 7. Shapley value ratio of models for OpenSARShip-1.0 dataset. From the outermost ring to the innermost ring, they represent A-ConvNet, AM-CNN, EfficientNet-B0, ResNet18, MVGGNet, and ConvNeXtTiny, respectively.

AM-CNN, EfficientNet-B0, ResNet18, MVGGNet, and ConvNeXtTiny. The contribution of the target region dominates the prediction results for A-ConvNet, AM-CNN, EfficientNet-B0, and ResNet18. Conversely, for MVGGNet and ConvNeXtTiny networks, the clutter region's contribution takes precedence. Similar to conclusions drawn from the MSTAR dataset, as the network parameter count increases, the network exhibits enhanced fitting capabilities for clutter regions.

Fig. 8 depicts the Shapley values ratio for individual classes of the OpenSARShip-1.0 dataset under different network architectures. Similar to the MSTAR experiments, due to the lack of comparability between Shapley values for different classes, they are normalized into ratio form. Negative Shapley ratios indicate a negative impact, suggesting that replacing the region with the proposed baseline values helps identify such instances, a finding validated in subsequent experiments.

For the BulkCarrier and Container classes, all six networks consistently exhibit a phenomenon where the contribution of the target region surpasses that of the clutter region. However, for the Cargo class, the clutter region's contribution to the network is more significant. Moreover, in the case of EfficientNet-B0, ResNet18, and MVGGNet, the target region shows a negative impact, indicating that not only does it fail to contribute to the output, but also it confuses the network's decision.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

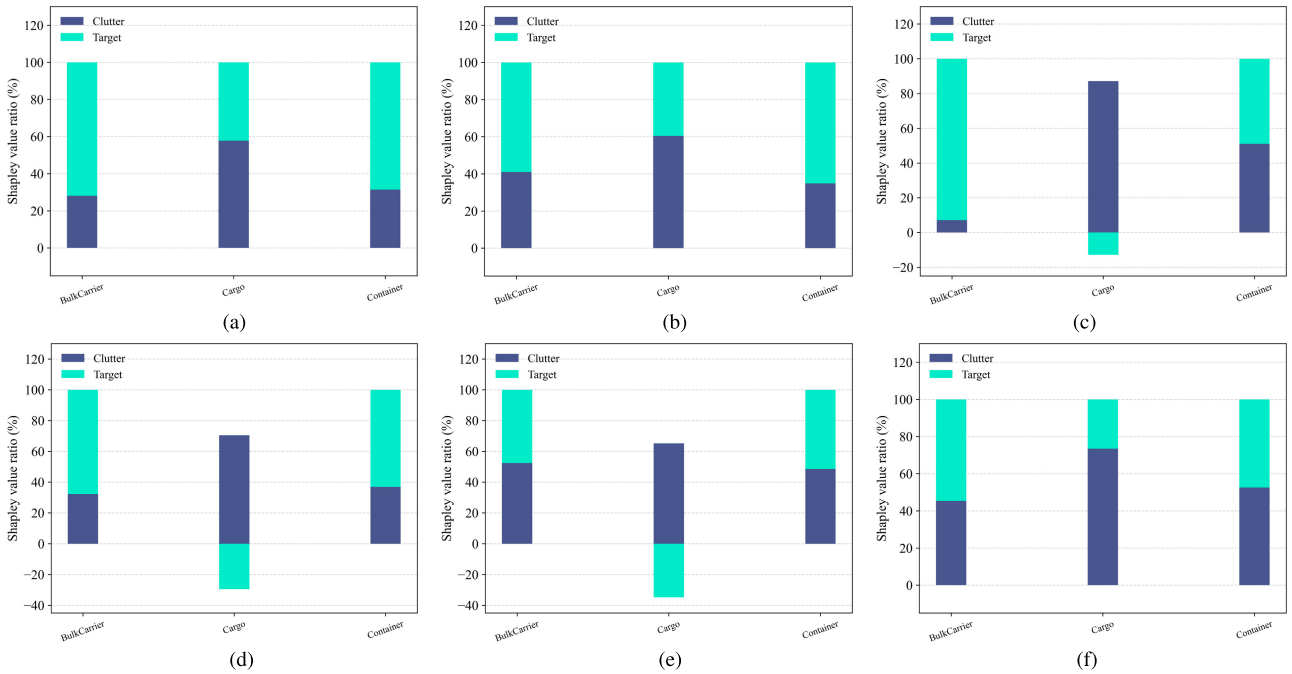


Fig. 8. Shapley value ratio of different DNNs for OpenSARShip-1.0. (a) A-ConvNet. (b) AM-CNN. (c) EfficientNet-B0. (d) ResNet18. (e) MVGGNet. (f) ConvNeXtTiny.

E. Different Baseline Values' Average Recognition Accuracy

It is usually unable to determine the true values of baseline values for real images, such as in the MSTAR dataset. To validate whether the methodology learns baseline values more accurately than random initialization or setting them to zero, the training and testing sets are initially partitioned based on the positive and negative contributions of the corresponding Shapley values. Regions corresponding to negative Shapley values, indicating masking, are masked, while regions corresponding to positive Shapley values are retained. A new training set and testing set, referred to as "After Masking" data, are obtained following the same partitioning scheme as outlined in Table IV, where specific regions corresponding to individual samples are selectively occluded. Subsequently, the network is thoroughly trained on the modified training set, and recognition accuracy is assessed on the testing set.

For instance, in the case of zero baseline values, SHAP method is employed to independently compute Shapley values for different regions of each sample. The regions associated with negative values are masked, those with positive values are retained, and samples with no masked regions remain unchanged. The aforementioned steps are repeated for zero, mean, random, and the proposed baseline values, yielding the results presented in Table VIII.

In the experiments, random baseline values are generated from a Rayleigh distribution with a mean of 0 and a variance of 1. When examining the methods of utilizing either a complete training set and partially occluded test set or vice versa, experiments unveiled a significant distribution shift between the test and training sets. This shift resembled more of a domain adaptation issue. Specifically, a substantial discrepancy in accuracy between the test and training sets was observed. For instance, employing a full training set and applying masking solely to the test set within A-ConvNet resulted in a training set accuracy reaching 97%, whereas the test set accuracy was merely 56%.

Experiments on six network models, including A-ConvNet, AM-CNN, and EfficientNet-B0, were conducted. "Before Masking" represents the standard training approach, serving as the control group. The values outside and inside the parentheses denote the recognition accuracy on the training and testing sets, respectively.

The recognition accuracy of the zero baseline values and mean baseline values methods are nearly identical, suggesting their potential to introduce additional irrelevant information, thus affecting the loyalty of computing Shapley values. Following closely, the random baseline values method demonstrates slightly better accuracy than the former two methods. Given its distribution follows a Rayleigh distribution, approximating the clutter of SAR backgrounds, it effectively initializes baseline values with certain prior knowledge. In contrast, the proposed method consistently outperforms the control group across all networks. This further validates that the baseline values obtained through the proposed method are closer to the actual values. The Shapley values computed based on the proposed method more faithfully reflect the contributions of different regions to the network's decision-making.

Through a comprehensive comparative analysis and detailed ablation study, the effectiveness of the proposed method has been quantitatively demonstrated. In this section, the t-stochastic neighbor embedding (SNE) algorithm [38] is employed to visualize the features learned by A-ConvNet before masking. Additionally, it visualizes the features learned

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII

ACCURACY UNDER DIFFERENT SHAPLEY VALUES OPTIMIZATION OF MODELS (UNIT IS %, WITHIN PARENTHESES ARE THE TRAINING ACCURACIES, AND OUTSIDE PARENTHESES ARE THE TESTING ACCURACIES)

Dataset	Model	Before Masking	After Masking			
			Zero Baseline values	Mean Baseline values	Random Baseline values	**Proposed**
MSTAR	A-ConvNet	97.73(94.29)	86.32(65.94)	88.94(67.41)	91.66(76.99)	**99.26(98.91)**
	AM-CNN	100(97.10)	89.10(71.62)	89.68(67.37)	92.21(76.08)	**100(98.84)**
	EfficientNet-B0	100(93.79)	86.79(64.44)	89.57(72.24)	91.07(76.65)	**100(97.91)**
	ResNet18	99.86(95.82)	87.01(66.56)	87.66(68.97)	90.19(78.15)	**100(98.50)**
	MVGGNet	99.97(94.04)	90.38(64.50)	86.54(63.97)	93.19(78.93)	**99.81(98.78)**
	ConvNeXtTiny	99.94(97.44)	87.58(63.66)	90.03(63.44)	92.48(80.11)	**100(97.87)**
OpenSARShip-1.0	A-ConvNet	77.08(69.17)	58.33(55.00)	59.38(56.67)	67.08(60.83)	**91.88(81.67)**
	AM-CNN	92.08(76.67)	65.42(50.83)	64.58(60.00)	71.88(60.83)	**94.38(85.83)**
	EfficientNet-B0	93.33(66.67)	66.46(47.50)	68.96(53.33)	72.71(62.50)	**96.04(91.67)**
	ResNet18	91.88(75.83)	67.29(46.67)	67.08(47.50)	72.08(60.00)	**94.38(87.5)**
	MVGGNet	92.29(86.67)	59.58(45.83)	65.83(44.17)	73.96(63.33)	**94.79(90.00)**
	ConvNeXtTiny	80.21(64.17)	60.21(52.50)	67.71(55.00)	70.21(61.67)	**96.04(89.17)**

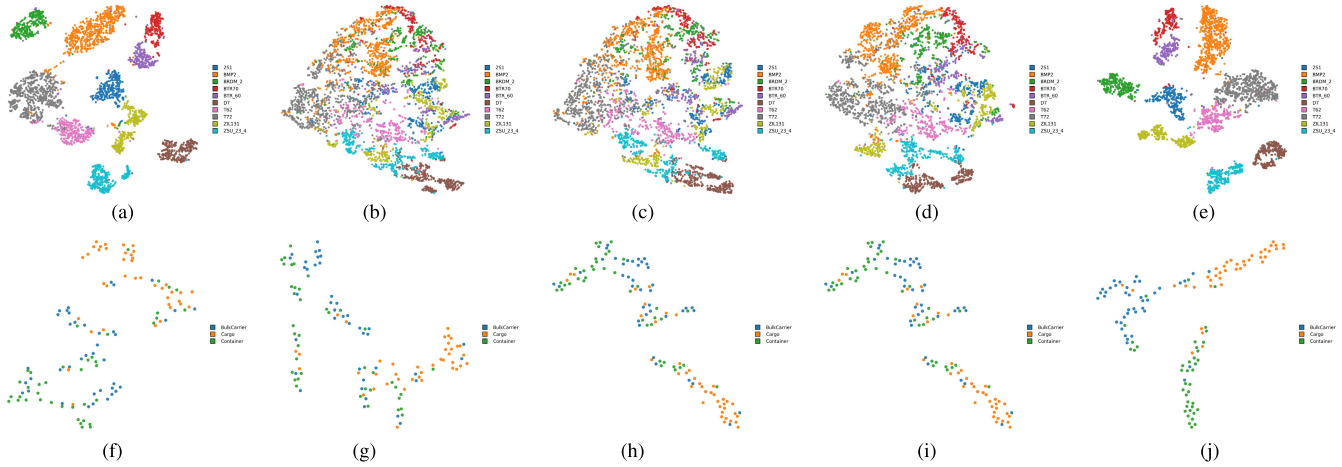


Fig. 9. t-SNE visualizations of the learned features with different baseline values. The upper row is for the MSTAR dataset and the lower row is for the OpenSARShip-1.0 dataset. (a) Before masking. (b) Zero baseline values. (c) Mean baseline values. (d) Random baseline values. (e) Proposed baseline values. (f) Before masking. (g) Zero baseline values. (h) Mean baseline values. (i) Random baseline values. (j) Proposed baseline values.

through the three masking methods and the proposed masking method mentioned in Table VIII. Fig. 9 presents the visualization results of the learned features corresponding to the five experimental settings outlined in Table VIII. For instance, Fig. 9(a) illustrates the visualization results of the high-dimensional features acquired by the network, corresponding to the first row and first column in Table VIII. From Fig. 9, it can be observed that, compared to Fig. 9(b)–(d), the boundaries between all types are more pronounced, indicating that the features in Fig. 9(e) are more separable.

V. CONCLUSION

In this article, a general algorithmic framework for SAR image classification tasks based on masking methods is proposed. Leveraging this framework, SHAP attributions are conducted on both the MSTAR and OpenSARShip-1.0 datasets to assess contributions from various regions. Furthermore, to accurately reflect the distinct impacts of different regions in SAR images on the overall classification outcome, a novel masking method, which refrain from introducing extra information and aim to eliminate causal information as much as possible, is proposed. The masking method based on multivariate interaction and SAR clutter distribution is presented for estimating Shapley values. This method of estimating optimal baseline values for Shapley value computation ensures the credibility of employing the SHAP method for attributions. Demonstrated through both qualitative and quantitative methods, the suitability of it as a faithful baseline value is established. The proposed masking method proves to be reliable and can serve as a faithful baseline value not only for SHAP but also for various other XAI methods.

Through SHAP attributions, the discovery reveals that the target region and background clutter region contain the majority of information required for classification. There exists correlation between the background and target classes in SAR data, and deep learning methods tend to overfit

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.

to the information from background clutter. Omitting background clutter regions severely impacts the performance of the network. The question whether DNNs can capture causal information embedded in background clutter, such as the geometric structure of the target and EM scattering information, still needs exploration. This suggests that DNNs may not make judgments based on the target itself as expected. Furthermore, under different imaging conditions, causal features (i.e., target features) are unstable. Current research on adversarial attacks indicates that deep learning is susceptible to minor variations in target features [39]. The robustness of deep learning in real-world settings may hinge upon its capacity to establish and exploit causal relationships. Consequently, there is a need to enhance this capability within deep learning methods to address uncertainties and adapt to changes in target features.

REFERENCES

[1] L. Liu et al., "Deep learning for generic object detection: A survey," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 261–318, Feb. 2020.

[2] X. Zhang, S. Feng, C. Zhao, Z. Sun, S. Zhang, and K. Ji, "MGSFA-Net: Multiscale global scattering feature association network for SAR ship target recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 4611–4625, 2024.

[3] W. Li, W. Yang, W. Zhang, T. Liu, Y. Liu, and L. Liu, "Hierarchical disentanglement-alignment network for robust SAR vehicle recognition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9661–9679, 2023.

[4] A. Hosein Oveis, E. Giusti, S. Ghio, G. Meucci, and M. Martorella, "LIME-assisted automatic target recognition with SAR images: Toward incremental learning and explainability," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 9175–9192, 2023.

[5] J. Fan, Z. Sui, and X. Wang, "Multiphysical interpretable deep learning network for oil spill identification based on SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–15, 2024, Art. no. 4202015, doi: 10.1109/TGRS.2024.3357800.

[6] M. Kohlbrenner, A. Bauer, S. Nakajima, A. Binder, W. Samek, and S. Lapuschkin, "Towards best practice in explaining neural network decisions with LRP," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–7.

[7] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," in *Proc. 34th Int. Conf. Mach. Learn.*, 2017, pp. 3145–3153.

[8] C. Belloni, A. Balleri, N. Aouf, J.-M. Le Caillec, and T. Merlet, "Explainability of deep SAR ATR through feature analysis," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 1, pp. 659–673, Feb. 2021.

[9] W. Li, W. Yang, L. Liu, W. Zhang, and Y. Liu, "Discovering and explaining the noncausality of deep learning in SAR ATR," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[10] A. Kapishnikov, T. Bolukbasi, F. Viegas, and M. Terry, "XRAI: Better attributions through regions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 4947–4956.

[11] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[12] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[13] M. T. Ribeiro, S. Singh, and C. Guestrin, "'Why should I trust you?' Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 1135–1144.

[14] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–10.

[15] H. Zhang, Y. Xie, L. Zheng, D. Zhang, and Q. Zhang, "Interpreting multivariate Shapley interactions in DNNs," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, 2021, vol. 35, no. 12, pp. 10877–10886.

[16] R. Weber, "Probabilistic values for games," in *The Shapley Value: Essays in Honor of Lloyd S. Shapley*. Cambridge, U.K.: Cambridge Univ. Press, 1988, pp. 101–119.

[17] J. Ren, Z. Zhou, Q. Chen, and Q. Zhang, "Can we faithfully represent absence states to compute Shapley values on a DNN?" in *Proc. 11th Int. Conf. Learn. Represent.*, 2022, pp. 1–30.

[18] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[19] D. Malmgren-Hansen and M. Nobel-Jørgensen, "Convolutional neural networks for SAR image segmentation," in *Proc. IEEE Int. Symp. Signal Process. Inf. Technol. (ISSPIT)*, Dec. 2015, pp. 231–236.

[20] I. Covert, S. M. Lundberg, and S.-I. Lee, "Understanding global feature contributions with additive importance measures," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 17212–17223.

[21] L. Merrick and A. Taly, "The explanation game: Explaining machine learning models using Shapley values," in *Machine Learning and Knowledge Extraction*, Dublin, Ireland. Cham, Switzerland: Springer, 2020, pp. 17–38.

[22] C. Agarwal and A. Nguyen, "Explaining image classifiers by removing input features using generative models," in *Proc. Asian Conf. Comput. Vis.*, 2020, pp. 1–17.

[23] C. Frye, D. de Mijolla, T. Begley, L. Cowton, M. Stanley, and I. Feige, "Shapley explainability on the data manifold," 2020, *arXiv:2006.01272*.

[24] K. M. Borgwardt, A. Gretton, M. J. Rasch, H.-P. Kriegel, B. Schölkopf, and A. J. Smola, "Integrating structured biological data by kernel maximum mean discrepancy," *Bioinf.*, vol. 22, no. 14, pp. 49–57, 2006.

[25] W. Deng et al., "Joint clustering and discriminative feature alignment for unsupervised domain adaptation," *IEEE Trans. Image Process.*, vol. 30, pp. 7842–7855, 2021.

[26] P. Dardouillet, A. Benoit, E. Amri, P. Bolon, D. Dubucq, and A. Credoz, "Explainability of image semantic segmentation through shap values," in *Proc. ICPR-XAIE*, 2022, pp. 188–202.

[27] J. Ren, M. Li, Q. Chen, H. Deng, and Q. Zhang, "Defining and quantifying the emergence of sparse concepts in DNNs," 2021, *arXiv:2111.06206*.

[28] J. C. Harsanyi, *A Simplified Bargaining Model for the N-Person Cooperative Game*. Osaka, Japan: Osaka University's Institute of Social, International Economic Review (IER), 1982.

[29] G. A. Lampropoulos, A. Drosopoulos, and N. Rey, "High resolution radar clutter statistics," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 35, no. 1, pp. 43–60, Jan. 1999.

[30] M. Amoon and G.-A. Rezai-Rad, "Automatic target recognition of synthetic aperture radar (SAR) images based on optimal selection of Zernike moments features," *IET Comput. Vis.*, vol. 8, no. 2, pp. 77–85, 2014.

[31] L. Huang et al., "OpenSARShip: A dataset dedicated to Sentinel-1 ship interpretation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 195–208, Jan. 2018.

[32] Z. Shao et al., "Scale in scale for SAR ship instance segmentation," *Remote Sens.*, vol. 15, no. 3, p. 629, 2023.

[33] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, "Target classification using the deep convolutional networks for SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4806–4817, Aug. 2016.

[34] J. Zhang, M. Xing, and Y. Xie, "FEC: A feature fusion framework for SAR target recognition based on electromagnetic scattering features and deep CNN features," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2174–2187, Mar. 2021.

[35] E. A. Li and W. J. Wang, "Research and exploration on the interpretability of deep learning model in radar image," *SCIENTIA SINICA Informationis*, vol. 52, pp. 1114–1134, Jun. 2022.

[36] R. Fjelland, "Why general artificial intelligence will not be realized," *Humanities Social Sci. Commun.*, vol. 7, no. 1, pp. 1–9, Jun. 2020.

[37] M. Zhang, J. An, D. H. Yu, L. D. Yang, L. Wu, and X. Q. Lu, "Convolutional neural network with attention mechanism for SAR automatic target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[38] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, pp. 2579–2605, 2008.

[39] B. Peng, B. Peng, J. Zhou, J. Xia, and L. Liu, "Speckle-variant attack: Toward transferable adversarial attack to SAR target recognition," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.



Zongyong Cui (Member, IEEE) received the B.E. and Ph.D. degrees in signal and information processing from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2007 and 2015, respectively.

From 2013 to 2014, he was a Visiting Student with the Department of Electrical and Computer Engineering, National University of Singapore, Singapore. He is currently an Associate Professor with the School of Information and Communication Engineering, UESTC. His research interests include radar image processing, target recognition, and machine learning.

Dr. Cui is a reviewer of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and *IET Radar, Sonar & Navigation*.



Zhiyuan Yang (Student Member, IEEE) received the B.E. degree in communication engineering from Zhengzhou University, Zhengzhou, China, in 2022.

He is currently pursuing the M.S. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, China. His research interests include machine learning and synthetic aperture radar (SAR) image processing.



Zheng Zhou (Graduate Student Member, IEEE) received the B.E. degree from Chengdu University of Technology, Chengdu, China, in 2019.

He is currently pursuing the Ph.D. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu. His research interests include synthetic aperture radar (SAR) image processing, target detection, and few-shot learning.



Liqiang Mou (Student Member, IEEE) received the B.E. degree from Chengdu University of Technology, Chengdu, China, in 2022.

He is currently pursuing the M.S. degree with the School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu. His research interests include synthetic aperture radar (SAR) image processing, target recognition, and SAR target multidimensional feature fusion.



Kailing Tang (Student Member, IEEE) received the B.E. degree in information engineering from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2022, where he is currently pursuing the M.S. degree with the School of Information and Communication Engineering.

His research interests include synthetic aperture radar (SAR) image processing and object recognition.



Zongjie Cao (Member, IEEE) received the B.E. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1999 and 2005, respectively.

From 2006 to 2008, he was a Post-Doctoral Researcher with the Communication and Information System Postdoctoral Center, University of Electronic Science and Technology of China (UESTC), Chengdu, China. In 2008, he joined the School of Electronic Engineering, UESTC. He is currently a Professor with the School of Information and Communication Engineering, UESTC. He has authored or coauthored over 100 articles. His current research interests include radar signal processing, synthetic aperture radar imaging, and target recognition.

Dr. Cao is a reviewer for several international journals and conferences, such as ISPRS JOURNAL OF PHOTOGRAMMETRY AND REMOTE SENSING, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, and Remote Sensing.



Jianyu Yang (Member, IEEE) received the B.S. degree from the National University of Defense Technology, Changsha, China, in 1984, and the M.S. and Ph.D. degrees from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 1987 and 1991, respectively.

He is currently a Professor with UESTC. His research interests include synthetic aperture radar and statistical signal processing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 22:55:33 UTC from IEEE Xplore. Restrictions apply.